Habib University
shaping futures

Enterprise Software Development

CS 318x NB
Blueprint Term – Spring 2025

Credit Hours and Pre-requisites

Course Credit Hours: 3

Pre-requisite Courses: CS 232/CE 324 - Operating Systems and CS 355/CE 373 - Database Systems

Students are expected to be familiar with the basic Linux command line interface usage and can install tools like Docker, Terraform, and Ansible by following instructions without much hand-holding.

Previous experience with programming full-stack software is expected.

Course Information

Start Date	End Date	Class Location	Meeting Time

Hardware/Software Prerequisites: A Macbook or a Linux PC is recommended. A Windows machine may be set up with Windows Subsystem for Linux or a Linux Virtual Machine (Debian/Ubuntu distribution is recommended).
Content Area: This course fulfills the requirements of a CS upper-level elective or a free elective and can be counted as a CS elective toward the CS minor. Enrollment in the course is subject to meeting the prerequisites.

Instructor Information

Instructor: Ateeb Ahmed
Office Location: TBD
Email: ateeb.ahmed@sse.habib.edu.pk
Office Hours: TBD

Course Description

This course gives an overview of modern software and system design practices in industry. The course is split into four modules.

> The first module covers various techniques for modeling complexity in software such that it is easy to change and maintain. It also introduces various techniques software can be designed and trade-offs one must take into account when choosing how to design it.
>
> The second module covers the modern approach to system administration: DevOps. DevOps is a collaborative approach which combines software development with system administration. It encourages performing IT operations using code, automating the toil and reducing the time it takes to deliver the software to production. Students will learn about containers, will be introduced to container orchestration, automated infrastructure provisioning and system configuration. We will also introduce how to quantify reliability of systems using SLOs.
>
> The third module discusses different kinds of data systems and their use cases. We will compare RDBMS systems with various NoSQL solutions in terms of their scalability. The course will also introduce data systems other than traditional databases such as caches, stream processors and time series.

In the fourth module, the course ties everything taught above with discussions about how to build scalable software. Architectures of various well known large scale software systems are discussed. Students will be taught to do back of the envelope calculations and come up with system designs based on requirements.

Course Aims

This course covers a breadth of technologies and design techniques used in the modern software industry. At the end of the course, a student will have knowledge about best practices related to software architecture, DevOps engineering, various data systems and their use cases and designing scalable systems.

Non-Goals

The course does not try to go in depth of any of the subjects it covers. A lot of topics covered in this course are covered with broad strokes with some related hands-on exposure but in favor of covering the breadth of technologies in use in modern software design, the course has to sacrifice depth. Students are encouraged to deep dive into subjects of their interests and the instructor will provide as much help as possible.

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

- Demonstrate understanding of layered architectures and separation of concerns in software systems to write clean and maintainable code.
- Apply knowledge of different data systems to select the best storage solution and scalability strategy for the problem in hand.
- Develop automation for deploying software to increase the speed with which software is delivered to the customers.
- Evaluate different factors involved in selecting technology stacks and design the systems which can scale up and down with usage.

Mode of Instruction

To every extent possible, this course will take place in person. We will meet twice a week for 75-minute lectures. In the unfortunate circumstance where we need to go online, relevant instructions will be shared accordingly. For that contingency, you should have a computer with an internet connection that can run the latest browser version and Zoom.

For in-class sessions, you are expected to bring a laptop connected to the internet for following along the workshops and attempting the quizzes.

Engagement & Participation Rules

Listen actively and attentively to engage in meaningful discussions.

Mobile phones and laptops can only be used for legitimate class activities, e.g. note-taking, completing an assigned task.

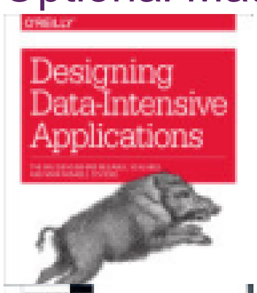
It is encouraged to bring laptops with Linux (WSL/VM or main OS) or Mac OS to the classroom for follow along workshops.

Leaving the class for more than 10 minutes would be considered an absence, and would be reflected in the attendance roster as such.

Required Texts and Materials

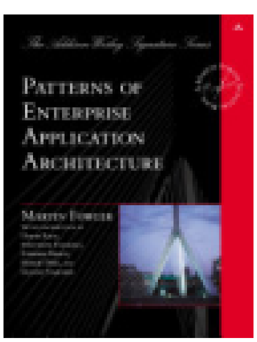
NA

Optional Materials



Designing Data-Intensive Applications
ISBN: 9781491903100
Authors: Martin Kleppmann
Publisher: "O'Reilly Media, Inc."
Publication Date: 2017-03-16

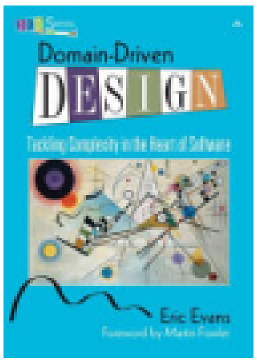
Design Patterns
ISBN: 9783827328243
Authors: Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides
Publisher: Pearson Deutschland GmbH
Publication Date: 1995-01-01



Patterns of Enterprise Application Architecture
ISBN: 9780133065213
Authors: Martin Fowler

Publisher: Addison-Wesley
Publication Date: 2012-03-09

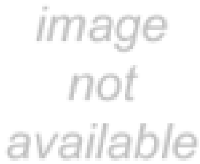
A Philosophy of Software Design
ISBN: 9781732102200
Authors: John Ousterhout
Publisher: Yaknyam Publishing
Publication Date: 2018-04-10



Domain-driven Design
ISBN: 9780321125217
Authors: Eric Evans
Publisher: Addison-Wesley Professional
Publication Date: 2004-01-01

Site Reliability Engineering
ISBN: 9781491951170
Authors: Niall Richard Murphy, Betsy Beyer, Chris Jones, Jennifer Petoff Publisher: "O'Reilly Media, Inc."
Publication Date: 2016-03-23

Grokking the System Design Interview
ISBN: 9798766433668
Authors: Design Gurus



Publication Date: 2021-12-18

Assessments

Assessment Type	Quantity	Weight (%)	Remarks
Projects	6	55	6 small projects aimed at giving some hands on exposure to technologies we cover in this course.
Weekly Quizzes	10	15	Quizzes will cover previous week's topic and will be taken during class.
Term/Final Exam	1	25	MCQs + Theoretical Questions
Case Study - Viva	1	5	Case Study Viva - In group

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)

B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Late Submission Policy

Students are encouraged to submit work much earlier than the deadline to have spare time in case of any mishaps at the last moments. If there is some problem, the instructor should receive an email about it at least an hour before the deadline so it can be remedied. Any communication after the deadline will not be entertained.

A student may submit only one of the assigned projects or report with a delay of 48 hours maximum without any penalty. It is recommended that the students only use this policy in case of an emergency. More than one late submission will not be accepted.

Week-Wise Schedule (Tentative)

Fall 2025 Weekly Schedule*

Weekly Schedule (Fall 2025)

Week	Dates	Topics	Assessments & Due Dates	Notes
1	Aug 19, 21	DDD, Layered & Service-Oriented Architectures, MVC, Domain Model ↔ Data Mapping	–	
2	Aug 26, 28	**API Fundamentals** (OpenAPI, gRPC, GraphQL, Circuit Breakers, Secrets Mgmt.)	Project 1 Released	
3	Sept 2, 4	Architectural Styles: Monoliths, Microservices, Modular, Case Studies	Quiz 1	Sept 4: 12th Rabi-ul-Awwal (Holiday – Thu class canceled)
4	Sept 9, 11	Scalability Techniques (CDNs, GeoDNS, Multi-AZ, Event-Driven)	Quiz 2; Project 1 Due; Project 2 Released	
5	Sept 16, 18	Continuous Integration & Continuous Delivery	Quiz 3	
6	Sept 23, 25	Containers & Kubernetes	Quiz 4; Project 2 Due; Project 3 Released	
7	Sept 30, Oct 2	Kubernetes & GitOps	Project 3 Due	Midterm week (no midterm for this course)
8	Oct 7, 9	Infrastructure Provisioning & Config Management	Quiz 5; Project 4 Released	Midterm week cont.
9	Oct 14, 16 (+Extra Session)	Site Reliability Engineering (SLOs, SLIs, SLAs)	Quiz 6	**Extra Thu session (Oct 16) covers material from Oct 30**
—	**Oct 11 (Sat)**	**Make-up Class for Tue Oct 28**	–	University Working Saturday
10	Oct 21, 23	Data Systems Overview, CAP Theorem	Quiz 7; Project 4 Due	
11	Oct 28, 30	**– No Classes (Rescheduled earlier) –**	–	Oct 28 → moved to Oct 11; Oct 30 → covered on Oct 16
12	Nov 4, 6	Scaling RDBMS & NoSQL Systems	Quiz 8; Project 5 Released (Term Exam (MCQs + Theory))	
13	Nov 11, 13	Event Sourcing, Stream Processing, Time Series DBs	Quiz 9	Nov 9: Iqbal Day (Holiday)
14	Nov 18, 20	Back-of-the-Envelope Calculations, Latency Numbers, QPS	Project 5 Due; Project 6 Released	Nov 18–21: Spring 2026 Enrollment
15	Nov 25, 27	System Design Case Study: YouTube / Netflix	Quiz 10	Nov 28–29: Conference Days
16	Dec 2, 4	System Design Case Study: Uber / WhatsApp	–	Dec 4: Spring 2026 Enrollment Opens; Dec 5: Last Class
—	Dec 6 (Sat), Dec 9 (Tue)	System Design Case Study: Google Drive / S3	Project 6 Due	Reading Days
Finals	Dec 10–13 & 15–16	TErm Exam (MCQs + Theory)	–	Dec 16: Incomplete Grade Petition Deadline

Notes:

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it deems necessary.

† Subject to the sighting of the new moon.
‡ No Class(es).

§ University's Examination Policy is available in the Academic Policies folder on the Faculty/Staff Portal.

Attendance Policy

Students are expected to maintain 100% attendance in the courses at HU. However, all students must maintain class attendance per the attendance threshold document shared by the ROto deal with any unforeseen situations at your end during the semester. If one cannot participate in any session, inform the instructor within 24 hours with a reason. Noncompliance will eventually lead to withdrawing/failing the student(s) from this course. Attendance will be marked manually in the class per the University's attendance policy.

Final Exam Policy

Final exam will consist of 60 MCQs to be answered in 60 minutes.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated,// bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.

b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.

c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Plagiarism Policy

All assessments are individual and collaboration is forbidden except when to the extent allowed by the instructor. You may learn something from external sources to solve projects but do not copy the solutions. All such sources should be cited. Plagiarism will not be tolerated.

Use of AI

The goal of assessments is not to test students but to reinforce the learning, therefore to get the most out of this course, students are encouraged to arrive at the solutions themselves and only when they have tried everything, use an AI. AI has become part of the modern software development process, therefore, this course will not penalize its use as long as it is cited. Remember, AI does hallucinate and produce wrong solutions. You are encouraged to be critical of anything an AI tells you and come up with your own solutions based on your intuition, AI output and any external references.

The report is for the students to consider all the tradeoffs and decisions that go into making a modern software system and critique and justify their use of technology. This goal is immediately negated when a student uses an AI, therefore use of AI is strictly forbidden for the report. The instructor has already used AI to generate reports with many prompts and has a good idea of what an AI influenced report looks like.

Program Learning Outcomes (For Administrative

Review) Upon graduation, students will have the following abilities:

PLO3: Analysis and Design: perform technical analysis and design using core computing and mathematical knowledge.

PLO4: Systems: apply the knowledge of computing systems.

PLO6: Problem Solving: identify and analyze problems and propose effective computing based solutions.

PLO7: Practical Exposure: make effective use of current tools, technologies, and good industry practices.

Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)	
	CLOs of the course are designed to cater following PLOs: PLO 3: Analysis and Design PLO 4: Systems PLO 6: Problem Solving PLO 7: Practical Exposure
	Distribution of CLO weightage for each PLO

	CLO 1	CLO 2	CLO 3	CLO 4
PLO 3	x			x
PLO 4		x		
PLO 6		x	x	
PLO 7	x	x	x	x

Mapping of Assessments to CLOs

Assessments	**CLO #01**	**CLO #02**	**CLO #03**	**CLO #04**
Domain Entity Mapping	x			
SDK Generation from API Spec	x			
Github Actions Pipeline		x		
Docker Image Creation		x		
Replicated RDBMS			x	
Report				x
Midterm	x	x		
Final		x	x	x
Quizzes 1-3	x			
Quizzes 4-7		x		
Quizzes 8-9			x	
Quiz 10				x

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

share their unique experiences, values and beliefs

be open to the views of others

honor the uniqueness of their colleagues

appreciate the opportunity that we have to learn from each other in this

community value each other’s opinions and communicate in a respectful manner

keep confidential discussions that the community has of a personal (or professional) nature

use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar’s Office.